1. Posterior sampling: why we did it

The purpose of this short presentation is to explain our posterior sampling study. Our question is: how sure are we about the model behind that forecast? A Bayesian model does not give us only one set of predictor weights. It gives us a distribution of plausible weights after seeing the data. That is called the posterior. A weight describes how a predictor is associated with the final grade while the other predictors are held fixed.

Think of sampling as taking many plausible versions of the fitted model, each consistent with its assumptions and the observed data. We can then ask whether an association is usually positive, whether it is large enough to matter, and how the uncertainty affects a forecast. We are drawing possible parameter values, not inventing additional students.

This short section explains the simulation checks in our earlier consolidated study. That fit used 148 earlier students. The weekly model uses 153 training students and different predictors, so the numerical checks I show are not new validation results for the weekly model.

2. Two ways to sample the same posterior

There are two ways we sampled the same posterior in this study. First, this model has an exact mathematical solution. We can draw directly from it. We took 20,000 independent draws, meaning that one draw does not depend on the previous draw.

Second, we ran MCMC. The letters mean Markov chain Monte Carlo. In plain language, it is a sequence of random updates designed to explore the posterior. A chain is one run of those updates. Successive draws can be related, unlike the independent draws on the left.

We used a method called Gibbs sampling: update the predictor weights given the current noise level, then update the noise level given those weights, and repeat. Noise means variation in final grades that the predictors do not explain.

Why do both? The exact solution is a reference answer for this model. Comparing Gibbs sampling with it gives us a strong numerical check and helps us understand sampling before considering models without exact solutions. The weekly forecast still uses the exact formula. MCMC did not replace it or make its predictions automatically better.

3. How Gibbs MCMC works

Here is what the Gibbs sampler actually did. We began four separate runs at different noise levels, rather than relying on a single starting point. In each run we repeatedly sampled predictor weights and then the unexplained variance. If we run long enough and the procedure is working, the retained draws should explore the same posterior regardless of where we started.

We discarded the first 1,000 updates from each chain as warm-up, so the initial settings would have less influence on our summary. We then kept 4,000 draws from each of the four chains: 16,000 retained MCMC draws in total. These are separate from the 20,000 independent direct draws on the previous slide.

The recorded experiment sampled the final-grade regression conditional on a fixed set of predictors. Missing predictor values had been filled using training-data medians, and that filled-in design stayed fixed during sampling. Therefore this experiment does not account for uncertainty in those missing values, and it is not joint sampling of every unknown in the full DAG. That matters when interpreting how complete the reported uncertainty is.

4. The sampling agreed with the exact solution

We did not just run the sampler and assume it worked. The chart shows part of the retained chains for one predictor weight and the unexplained variance. The colored lines overlap and move around a common region. That is a useful visual check, but it is not enough on its own.

R-hat compares variation within and between chains. Values near one indicate that the runs are behaving similarly; the largest value here was 1.0007. Effective sample size accounts for the fact that linked MCMC draws repeat information. The smallest bulk effective sample size was 12,194. This is an effective number of parameter draws, not a number of students.

We also compared the sampled parameter means with the exact mathematical means. The largest difference was about 2.01 Monte Carlo standard errors: the estimated numerical uncertainty caused by taking a finite number of draws. Together, these checks support agreement with the exact posterior for this fitted model. They do not prove that the model is correct, that a predictor is causal, or that future grade forecasts will be accurate.

Source: consolidated presentation slide 5; posterior_study/analysis.py and posterior_results/provenance.json. The trace displays the first 800 retained updates per chain.

5. What the posterior tells us about the inputs

On the previous slide, we checked that the sampler was exploring the posterior properly and agreeing with the exact solution. Now we can ask the more useful question: what does that posterior tell us about the student inputs?

Let me walk through one row rather than read the whole table. For LMS page views, the average estimated association is plus 0.34 final-grade points. That is for an increase of one standard deviation in page views—roughly one typical amount of variation in our training data—while holding grades and the other predictors fixed.

But 0.34 is only the center of the distribution. The 95% credible interval runs from minus 0.21 to plus 0.88. Given this model and its data assumptions, that interval contains 95% of the posterior probability for this weight. It includes both slightly negative and positive values, so the direction is uncertain. The interval is about the predictor's weight; it is not a range for a student's final grade.

The same pattern appears across all six rows: the estimated associations are small, and every interval includes zero. In this fit, each predictor also has less than a 1% posterior probability of having an absolute weight above one final-grade point per standard deviation. That is stronger evidence about small magnitude than simply observing that the intervals cross zero.

So the message is that, once the included grades and other predictors are accounted for, these inputs have small, uncertain additional associations in this particular model. It does not mean wellness or engagement are unimportant in students' lives, or that changing them would have no effect.

These numbers come from the earlier 148-student fit, not the weekly model's 153-student training fit. The table uses the analytic posterior; MCMC was our check that sampling could reproduce it.

That brings us to the final slide: understanding a predictor's uncertainty is one achievement. Knowing whether the model makes useful forecasts is a separate question.

Source for questions: consolidated presentation slide 9; posterior_results/coefficient_posteriors.csv.

6. What this establishes—and what it leaves open

The table we just saw tells us how uncertain we are about the fitted associations. To close, I want to separate three questions that can otherwise sound like the same thing.

First: how sure are we about a predictor's weight? That is what the credible intervals on the previous slide answer. They describe the model's uncertainty about an association after seeing the data.

Second: how sure are we about one student's final grade? That needs a prediction interval. Even if we knew every predictor weight exactly, students with similar observed evidence could still finish with different grades. A prediction interval therefore includes unexplained outcome variation as well as uncertainty about the fitted model. Taking more MCMC draws reduces numerical sampling error; it does not remove that real uncertainty.

Third: does the model forecast well for later students? We cannot answer that from a coefficient table or from chains that mix well. We have to compare forecasts with actual outcomes for students who were not used to fit the model, and compare against a reasonable baseline. That is the role of the separate prediction tests.

This explains why we did the simulation. The exact solution gave us a reference answer, and the Gibbs chains reproduced it. That supports our numerical implementation and helps us understand the posterior. It does not establish predictive usefulness or a causal benefit from coaching. The weekly model can continue to use the exact formula; it does not need MCMC to produce its forecasts.

One scope limit matters here: the earlier study kept filled-in predictor values fixed, so its uncertainty does not include uncertainty about those missing inputs. Its diagnostics also apply to that fitted model, not automatically to the separate weekly fit.

My takeaway is: we checked the posterior calculation, used it to describe the uncertain associations, and kept the practical question—how well we predict later students—for a separate test.

References for questions: consolidated presentation slides 4, 5, 9 and 17; posterior_study/analysis.py; posterior_results/provenance.json; Vehtari et al. (2021), DOI 10.1214/20-BA1221.